



2004
FORM VI
TRIAL HSC EXAMINATION

Chemistry

General Instructions

- Working time – 3 hours
- Board-approved calculators may be used
- Write using blue or black pen
- Draw diagrams using pencil
- A Data Sheet and Periodic Table are provided at the back of this paper
- Write your student number at the top of pages 7, 11, 15 and 19

CHECKLIST	
Each boy should have the following:	
1 Question Paper	
1 Multiple Choice Answer Sheet	
1 4-page Writing Booklet	

Total marks (100)

Section I Pages 2 - 20

This section has two parts, Part A and Part B

Part A

Total marks (15)

- Attempt Questions 1 - 15
- Allow about 30 minutes for this Part

Part B

Total marks (69)

- Attempt Questions 16 - 29
- Allow about 2 hours for this Part

Section II Pages 21 - 28

Total marks (16)

- Attempt ONE Question from Questions 30 - 34
- Allow about 30 minutes for this Section

1 - MMB	2 - AKBB	3 - JAG	
4 - AKBB	5 - PRT	6 - JAG	7 - EPC

Part A**Total marks (15)****Attempt ALL Questions****Allow about 30 minutes for this Part**

Use the multiple-choice Answer Sheet.

Select the alternative A, B, C or D that best answers the question. Fill the response circle completely.

Sample

$2 + 4 =$

(A) 2

(B) 6

(C) 8

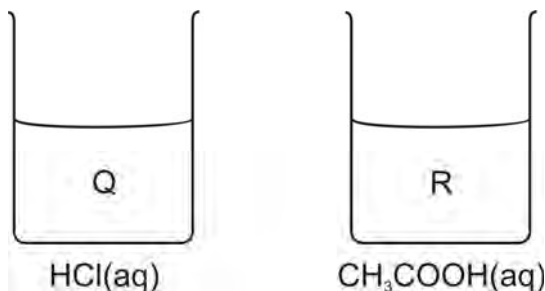
(D) 9

☐ (A)☒ (B)☐ (C)☐ (D)

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

☒ (A)☒ (B)☐ (C)☐ (D)If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.☒ (A)☒ (B)*correct*☐ (C)☐ (D)

1



Solution Q is a solution of hydrochloric acid (pH = 2.5), while solution R is a solution of acetic (ethanoic) acid (pH = 2.5).

Based on the above information and your knowledge of acids, which of the following statements is correct?

- (A) Solution R is stronger than solution Q.
- (B) Solution R is more concentrated than solution Q.
- (C) Solution Q is more concentrated than solution R.
- (D) Solution Q contains more H⁺(aq) than solution R.

2 Which of the following statements about ozone is valid?

- (A) Ozone depletion occurs only in the atmosphere above the South Pole.
- (B) Ozone is a vital gas in the stratosphere.
- (C) Ozone is a linear molecule.
- (D) Ozone is destroyed only by chlorofluorocarbons.

3 Which of the following types of radiation is the most penetrating?

- (A) α
- (B) β
- (C) γ
- (D) ${}^1_0\text{n}$

4 At what point is equilibrium reached in a reversible reaction?

- (A) When reactants stop changing into products.
- (B) When the molar concentrations of reactants and products are constant.
- (C) When the molar concentrations of reactants and products are equal.
- (D) When the activation energy of the forward and backward reactions are the same.

- 5 A boy wished to classify lemon juice according to its acid/base characteristics. To do this he diluted some lemon juice and then added three drops of bromothymol blue. What colour would you expect this indicator to be in dilute lemon juice?
- (A) Red
(B) Yellow
(C) Blue
(D) Colourless
- 6 Which of the following is the most common anode in commercial primary galvanic cells?
- (A) Zinc
(B) Mercury(II) oxide
(C) Manganese dioxide
(D) Lead
- 7 Which of the following pairs of aqueous solutions will produce a precipitate on mixing?
- (A) Sodium chloride and potassium nitrate
(B) Lead(II) chloride and potassium nitrate
(C) Potassium carbonate and barium nitrate
(D) Copper(II) sulphate and sodium chloride
- 8 What is the common name for 2-hydroxypropane-1,2,3-tricarboxylic acid?
- (A) Acetic acid
(B) Hydrochloric acid
(C) Sulphuric acid
(D) Citric acid

9 Consider the following data:

Half-reaction	$E^\circ_{\text{red}} / \text{V}$
$\text{W}^+ + \text{e}^- \rightarrow \text{W}$	2.3
$\text{X}^{3+} + \text{e}^- \rightarrow \text{X}^{2+}$	0.7
$\text{Y}^{2+} + 2\text{e}^- \rightarrow \text{Y}$	-0.7
$\text{Z}^{2+} + 2\text{e}^- \rightarrow \text{Z}$	-1.7

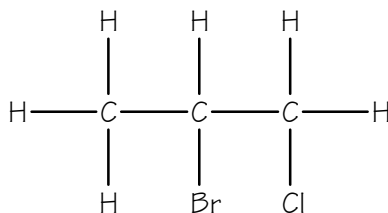
Using the data above, which of the following is the best reducing agent?

- (A) W
 - (B) W^+
 - (C) Z^{2+}
 - (D) Z
- 10 Which of the following needs to be monitored by industrial chemists working in coal-fired power stations?
- (A) Electricity generated by the station.
 - (B) Rate of formation of ammonia from its elements.
 - (C) Emission of steam and carbon dioxide.
 - (D) Emission of carbon monoxide and sulphur dioxide.
- 11 Which of the following will affect the amount of hydrogen iodide gas present at equilibrium in this reaction?
- $$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$$
- (A) Adding a catalyst.
 - (B) Adding an inert gas.
 - (C) Increasing the pressure.
 - (D) Increasing the temperature.
- 12 Which of the following is the most commercially significant addition polymer?
- (A) Nylon
 - (B) PVC
 - (C) PET
 - (D) Starch

13 Which of the following statements about neutralization is correct?

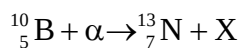
- (A) Neutralization is an electron transfer and is endothermic.
- (B) Neutralization is an electron transfer and is exothermic.
- (C) Neutralization is a proton transfer and is endothermic.
- (D) Neutralization is a proton transfer and is exothermic.

14 What is the IUPAC name of the following compound?



- (A) 2-bromo-3-chloropropane
- (B) 1-chloro-2-bromopropane
- (C) 2-bromo-1-chloropropane
- (D) 2-chloro-2-bromopropane

15 In the nuclear transformation below, what is X?



- (A) An electron
- (B) A proton
- (C) A neutron
- (D) A positron

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Class

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Student Number

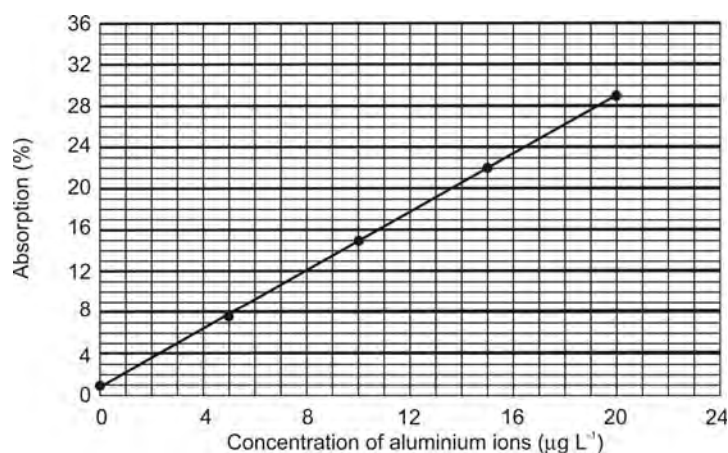
Part B**Total marks (69)****Attempt ALL Questions****Allow about 2 hours for this Part**

Answer the questions in the spaces provided

Show all relevant working in questions involving calculations

Marks**Question 16 (2 marks)**

Atomic absorption spectroscopy (AAS) can be used as an analytical tool for finding the concentration of elements in the ppm range. The graph below shows the relationship of absorption against concentration of aluminium ions.

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Use this graph to determine the Al^{3+} concentration in ppm for a sample which registered an absorption of 10%.

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Marks**Question 17** (6 marks)

The equation $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ represents the synthesis of ammonia from its component gases and is known as the Haber process.

- (a) Describe the geo-political conditions under which Haber developed the industrial synthesis of ammonia and evaluate its significance at this time in world history. **3**

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- (b) Explain why the Haber process is based on a delicate balancing act involving reaction energy, reaction rate and equilibrium. **3**

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Marks**Question 18** (5 marks)

Atomic absorption spectroscopy (AAS) is an extremely useful tool in the detection of metal ion concentrations.

- (a) Explain why AAS is of little use in identifying unknown substances. **3**

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- (b) Explain how AAS has had a major impact on the scientific understanding of the effects of trace elements. **2**

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Marks**Question 19** (6 marks)

The fermentation of glucose is a chemical process which has been known to humans for at least 5 thousand years.

(a) Write a chemical equation to represent the fermentation of glucose. **2**

(b) Under what physical conditions is fermentation optimised? **1**

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(c) One of the products of the fermentation process is frequently used as a solvent for both polar and non-polar solutes. Account for ethanol's ability to do this. **3**

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Section I – Part B (continued)

Class							
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Marks**Question 20** (5 marks)

Fossil fuels, which at present make up the bulk of the raw material used in the plastics industry, are a finite resource and likely to become severely depleted in the near future. Biopolymers have been suggested as a possible replacement for the petrochemicals produced from fossil fuels.

- (a) Cellulose is often considered the most useful compound from which to produce biopolymers. Describe the structure of cellulose. **2**

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- (b) (i) Identify a biopolymer which has recently been developed or is in the process of being developed, for commercial use. **1**

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- (ii) Name the specific enzyme or organism used to synthesise this biopolymer. **1**

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- (c) Suggest one benefit (apart from their renewability), of using biomass to produce polymers. **1**

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Marks**Question 21** (3 marks)

(a) Draw electron dot diagrams to show:

(i) an oxygen molecule.

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(ii) an ozone molecule.

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(b) State the difference in stability of ozone gas and oxygen gas.

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Marks**Question 22** (3 marks)

Esters are produced by reaction of an alkanoic acid and an alcohol.

- (a) Name a straight-chained alkanoic acid. **1**

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- (b) Name a primary alcohol. **1**

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- (c) Name the ester that would be produced by refluxing this acid with this alcohol. **1**

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Marks**Question 23** (7 marks)

Over time, the definitions of acids and bases have been refined. Using the historical development of ideas about acids, evaluate how advances in scientific understanding changed the direction of scientific thinking.

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Section I – Part B (continued)

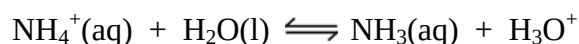
Class									
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Marks

Question 24 (5 marks)

A student was investigating the acid/base nature of salts, by adding the dry solid salts one at a time to water and then testing their pH.

When he did this with ammonium chloride, he noted that the $\text{pH} < 7$, and assumed that the following action had occurred.



- (a) Why does the above equation illustrate a Brønsted-Lowry acid, rather than an Arrhenius acid? 2

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- (b) From the above equation, give one example of an acid and its conjugate base, respectively. 1

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- (c) Briefly outline how you would perform a first-hand investigation to determine the concentration of an acidic substance using a computer-based technology. 2

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Marks**Question 25** (3 marks)

- (a) Identify two metallic ions which are found in hard water. **1**

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- (b) Describe a simple method of determining the hardness of water in a school laboratory. **2**

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Marks**Question 26** (6 marks)

While we usually think of the air around us as neutral, the atmosphere naturally contains acidic oxides of carbon, nitrogen and sulfur.

- (a) (i) Describe, using an equation, an example of a chemical reaction which releases sulfur dioxide. **2**

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- (ii) Identify a natural source of sulfur dioxide. **1**

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- (b) (i) Describe, using an equation, an example of a chemical reaction which releases an oxide of nitrogen. **2**

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- (ii) Identify a natural source of nitric oxide (NO), a gas that is capable of destroying ozone, and is involved in the production of photochemical smog. **1**

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Section I – Part B (continued)

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Marks**Question 27** (4 marks)

- (a) Discuss the conditions under which nuclei are stable. 2

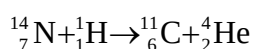
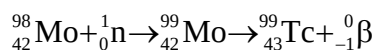
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- (b) The two equations below represent the formation of significant artificial isotopes: 2



Tc-99 is the most widely used radioactive isotope for diagnostic studies in nuclear medicine. C-11 is incorporated into organic compounds and used as a tracer in positron emission tomography (PET).

Discuss the production of commercial isotopes using these and / or other relevant examples.

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Marks**Question 28** (6 marks)

Galvanic cells were constructed using the metals A – E and the voltages measured under standard conditions. The results are shown in the table below.

Cell reaction	$E^\circ_{\text{cell}} / \text{V}$
$A + B^{2+} \rightarrow A^{2+} + B$	0.98
$B + D^{2+} \rightarrow B^{2+} + D$	1.05
$2C + B^{2+} \rightarrow 2C^+ + B$	1.68
$B + B^{2+} \rightarrow B^{2+} + B$	0.00
$B + E^{2+} \rightarrow B^{2+} + E$	0.66

- (a) Draw a labelled diagram of one of the cells used and identify clearly the reference cell.

3

- (b) Explain what is meant by standard conditions.

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Question 28 continued on page 21

Marks

Question 28 (continued)

(c) Construct a table of standard (half-cell) potentials from the data collected. **1**

(d) (i) Identify the best reducing agent. **1**

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(ii) Identify the best oxidising agent.

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Marks**Question 29** (8 marks)

Polyethylene is a chemical which has been of significant commercial importance in the past fifty years.

- (a) Outline the major steps in the industrial production of polyethylene, from the raw material used, to the finished product. **3**

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- (b) Many commercial polymers are produced by the modification of ethene molecules, such that a hydrogen is replaced by a side group, followed by a polymerisation reaction.
- (i) Identify one such “modified ethene” monomer, either by its common or systematic name, and using complete structural formula, write an equation to represent the polymerisation reaction, using **three** monomer units. **3**

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Question 29 continued on page 23

Marks

Question 29 (continued)

- (ii) Describe a use for the polymer you have identified, in part (i), in terms of its physical or chemical properties.

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Section II**Total marks (16)****Attempt ONE question from Questions 30 - 33****Allow about 30 minutes for this Section**

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 30	Industrial Chemistry 27
Question 31	Shipwrecks and Salvage
Question 32	Biochemistry of Movement
Question 33	Chemistry of Art 28-29

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Marks

Question 30 - Industrial Chemistry (16 marks)

- (a) Industrial chemists have researched and developed replacements for some natural products.
- (i) Identify one dwindling natural resource that is not a fossil fuel. **1**
 - (ii) Name a material that has been manufactured to replace the natural product identified in part (i). **1**
 - (iii) Explain why this replacement material is now manufactured. **1**
- (b)
- (i) Describe the use of sulphuric acid as a dehydrating agent. **1**
 - (ii) Explain how sulphuric acid may be used as an oxidant. **2**
- (c) Phosgene, or carbonyl chloride, COCl_2 , is a colourless, poisonous gas used in the production of some polymers. Carbonyl chloride decomposes as shown in the following equation.
- $$\text{COCl}_2(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{Cl}_2(\text{g})$$
- 1.00 mol of carbonyl chloride was placed in a 10.0 L sealed flask at 1250°C . At equilibrium 0.20 mol of carbonyl chloride was present in the flask. Calculate the value of the equilibrium constant for the decomposition of carbonyl chloride at 1250°C .
- (d) Chemistry laboratories buy 18M (concentrated) sulphuric acid and dilute this so that they are able to make the concentrations needed for day-to-day analysis. **3**
- Explain how you would **safely** dilute 18M sulphuric acid to make 2M sulphuric acid. Include safety precautions.
- (e) Sulfuric acid is such an important chemical in industry that its annual production may be used as an index of a nation's industrial activity. **4**
- Explain why sulphuric acid is such an important industrial chemical using three different industrial uses of sulphuric acid.

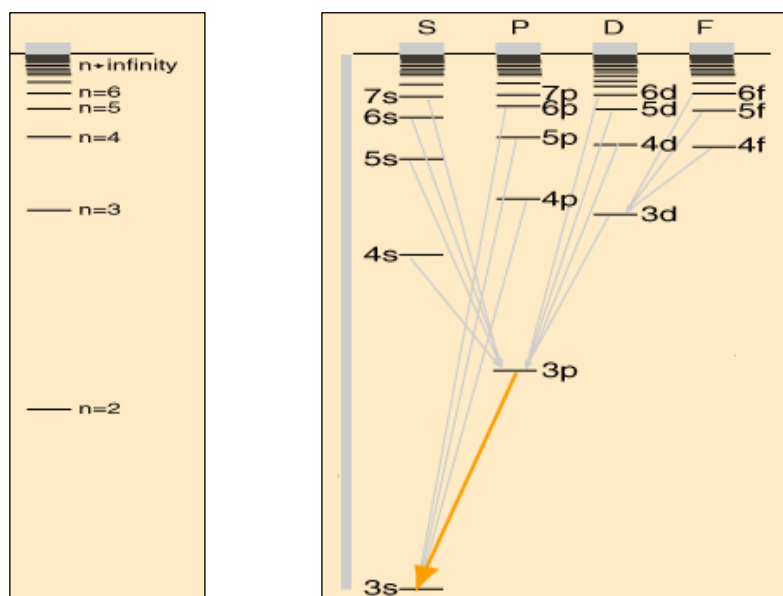
End of Question 30

Marks**Question 33 - Chemistry of Art (16 marks)**

- (a) Modern cosmetics are carefully formulated to be beneficial to the skin, or at least not harmful, but this was not always the case. Some of the pigments used in ancient Egyptian, Greek and Roman make-up are given below.

Face make-up	White lead	$2\text{PbCO}_3 \cdot \text{Pb(OH)}_2$
Lipstick	Cinnabar	HgS
Eye-shadow	Orpiment	As_2S_3
Mascara	stibnite	Sb_2S_3

- (i) What is the modern systematic name for orpiment? **1**
- (ii) What are some of the safety hazards associated with these pigments? **2**
- (b) (i) Identify the components of a paint. **1**
- (ii) Outline the processes and chemistry involved to prepare and attach pigments to surfaces in a named example of a medieval or earlier artwork. **2**
- (c) Explain the relationship between UV/visible absorption and reflectance spectra. **3**
- (d) Explain the main features of atomic absorption and emission spectra, making reference to the energy level diagrams for sodium and hydrogen (see below). **3**



Question 33 continues on page 29

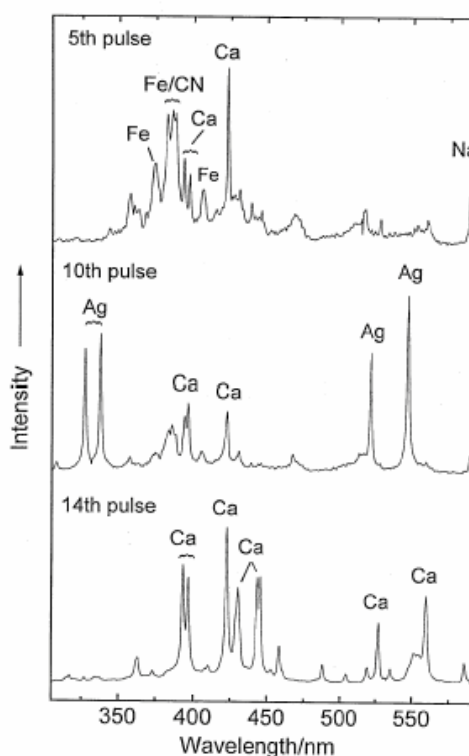
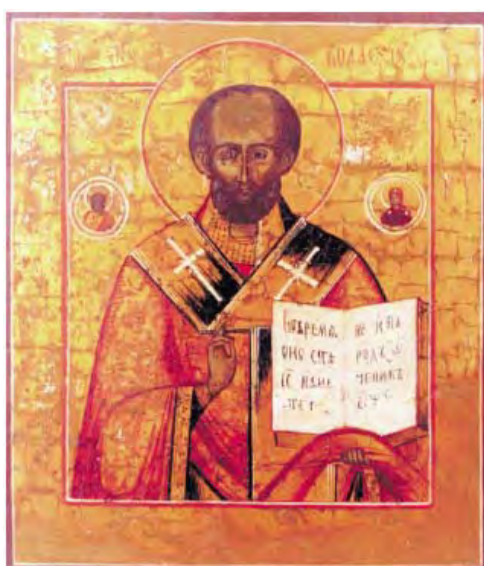
Marks

Question 33 (continued)

- (e) In laser microspectral analysis (LMA) a high energy laser pulse vaporises a minute amount of the material. Consecutive pulses dig deeper and deeper into the artwork, so that depth profiling is possible. The technique is very sensitive, using samples as small as 10^{-7} g. It may also be coupled with other techniques that can identify the individual pigments.

4

Russian icon of St Nicholas



The results of one LMA experiment on a nineteenth century Russian icon are shown above. A brown pigment in the paint is separated from the white ground by a metallic layer; the backing is wood.

Analyse the results and suggest compositions for the components of the three layers. Justify your answer.

End of Question 33

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Chemistry

Data Sheet

Avogadro's constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0 °C (273 K)	22.71L
at 25 °C (298K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m\Delta T$$

Standard Potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}_{(\text{s})}$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}_{(\text{s})}$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}_{(\text{s})}$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}_{(\text{s})}$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}_{(\text{s})}$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}_{(\text{s})}$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}_{(\text{s})}$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2} \text{H}_{2(\text{g})} + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}_{(\text{s})}$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}_{(\text{s})}$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}_{(\text{s})}$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}_{(\text{s})}$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}_{(\text{s})}$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2} \text{H}_{2(\text{g})}$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_{2(\text{g})} + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}_{(\text{s})}$	0.34 V
$\frac{1}{2} \text{O}_{2(\text{g})} + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}_{(\text{s})}$	0.52 V
$\frac{1}{2} \text{I}_{2(\text{s})} + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2} \text{I}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}_{(\text{s})}$	0.80 V
$\frac{1}{2} \text{Br}_{2(\text{l})} + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2} \text{Br}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2} \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2} \text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2} \text{H}_2\text{O}$	1.36 V
$\frac{1}{2} \text{Cl}_{2(\text{g})} + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2} \text{Cl}_{2(\text{aq})} + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2} \text{F}_{2(\text{g})} + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen	KEY										2 He 4.003 Helium												
3 Li 6.941 Lithium	4 Be 9.012 Beryllium	Atomic Number		79 Au 197.0 Gold		Symbol of element		8 O 16.00 Oxygen		9 F 19.00 Fluorine													
		Atomic Weight		Name of element																			
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium	13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon																
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.39 Zinc	31 Ga 69.72 Gallium	32 Ge 72.61 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton						
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.94 Molybdenum	43 Tc [98.91] Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon						
55 Cs 132.9 Cesium	56 Ba 137.3 Barium	57–71 Lanthanides		72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.8 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po [210.0] Polonium	85 At [210.0] Astatine	86 Rn [222.0] Radon					
87 Fr [223.0] Francium	88 Ra [226.0] Radium	89–103 Actinides		104 Rf [261.1] Rutherfordium	105 Db [262.1] Dubnium	106 Sg [263.1] Seaborgium	107 Bh [264.1] Bohrium	108 Hs [265.1] Hassium	109 Mt [268] Meitnerium	110 Uun — Ununilium	111 Uuu — Unununium	112 Uub — Ununbium	113 — — Ununtrium	114 Uuq — Ununquadium	115 — — Ununpentium	116 Uuh — Ununhexium	117 — — Ununseptium	118 Uuo — Ununoctium					

PERIODIC TABLE OF THE ELEMENTS

Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
--------------------------------	-----------------------------	-----------------------------------	--------------------------------	-----------------------------------	-------------------------------	-------------------------------	---------------------------------	------------------------------	---------------------------------	------------------------------	-----------------------------	------------------------------	--------------------------------	-------------------------------

Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
---------------------------------	------------------------------	-----------------------------------	-----------------------------	----------------------------------	----------------------------------	----------------------------------	-------------------------------	----------------------------------	------------------------------------	------------------------------------	---------------------------------	-------------------------------------	----------------------------------	------------------------------------

Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.



2004
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

CRIB							
Class							
Student Number							

Chemistry
Section I Part A
ANSWER SHEET

General Instructions

- Write your class and candidate number in the space provided.
- Attempt all questions 1 – 15
- Use a blue or black pen
- Select the alternative A, B, C, or D that best answers the question.
- Fill in the response circle completely.

					<u>% correct</u>
1.	(A)	(B)	(C)	(D)	72
2.	(A)	(B)	(C)	(D)	94
3.	(A)	(B)	(C)	(D)	97
4.	(A)	(B)	(C)	(D)	74
5.	(A)	(B)	(C)	(D)	57 *
6.	(A)	(B)	(C)	(D)	79
7.	(A)	(B)	(C)	(D)	54 *
8.	(A)	(B)	(C)	(D)	98
9.		(B)	(C)	(D)	
10.	(A)	(B)	(C)	(D)	78
11.	(A)	(B)	(C)	(D)	67
12.	(A)	(B)	(C)	(D)	
13.	(A)	(B)	(C)	(D)	69
14.	(A)	(B)	(C)	(D)	69
15.	(A)	(B)	(C)	(D)	74

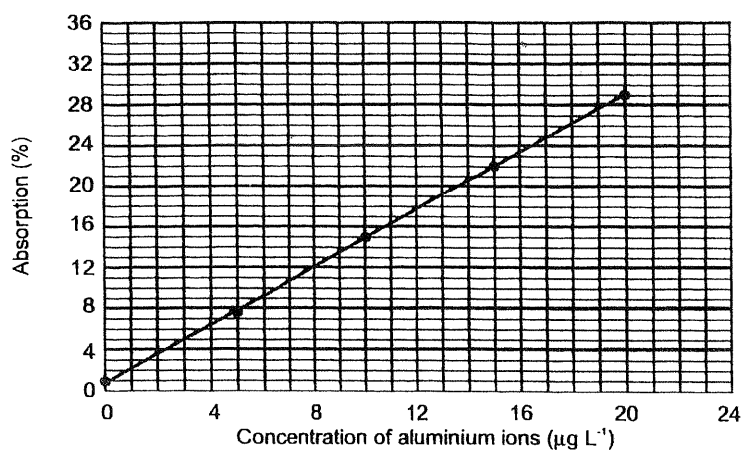
PART B

Marks

Question 16 (2 marks)

Atomic absorption spectroscopy (AAS) can be used as an analytical tool for finding the concentration of elements in the ppm range. The graph below shows the relationship of absorption against concentration of aluminium ions.

2



Use this graph to determine the Al^{3+} concentration in ppm for a sample which registered an absorption of 10%.

Marks

16

Al^{3+} concⁿ 6.5 $\mu\text{g L}^{-1}$

1

or $6.5 \times 10^{-3} \text{ ppm}$

1

If this Al^{3+} concⁿ were misread from the graph, but logical value in ppm $\rightarrow$ 1 mark.

N.B Boys often wrote $[\text{Al}^{3+}]$ for the concentration of Al^{3+} .

Question 17 (6 marks)

The equation $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ represents the synthesis of ammonia from its component gases and is known as the Haber process.

- (a) Describe the geo-political conditions under which Haber developed the industrial synthesis of ammonia and evaluate its significance at this time in world history.

3

17 (a) Just prior to WWI Germany realised it was losing its access to natural fertilisers eg saltpetre, guano. Then Allies' naval blockade prevented importation of saltpetre from S America & so Germany needed to develop production of ammonia → 'home production' of fertilisers and explosives.
As the Haber process → NH_3 → prolonged WWI as Germans had synthetic fertilisers for their food production + NH_3 used in making explosives.

Marks
3

Good description

3

Reasonable description

2

V. brief description

1

Wrong world war (eg WW II)

-1

(importing NH_3 , rather than nitrogen compounds)

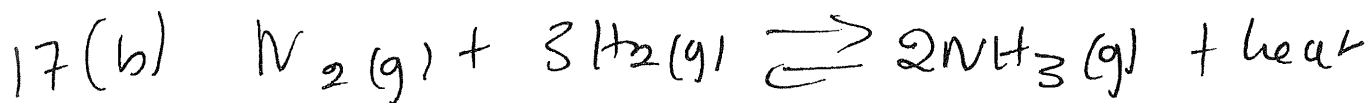
-1

17

(b)

Explain why the Haber process is based on a delicate balancing act involving reaction energy, reaction rate and equilibrium.

3



Forward reaction is exothermic

$\therefore \underline{T \uparrow} \rightarrow$ rate of reaction $\uparrow$, but will decrease yield of NH_3 (Le Chatelier's principle)

Catalyst $\rightarrow$ lower $E_a \rightarrow$ $\rightleftharpoons$ position reached faster, reasonable reaction rate.

(pressure) $\underline{P \uparrow}$ favours NH_3 yield (Le Chatelier's principle)

Removal of NH_3 also 'pulls' reaction to r.h.s (again Le Chatelier's)

Marks

Logical discussion.

including T , catalyst, P or removal of NH_3 and idea of compromise $\rightarrow$ 3 marks

Reasonable discussion.

- $T \uparrow$ and catalyst or temp and removal of NH_3 + some idea of compromise $\rightarrow$ 2 marks

Some points describing the Haber process

1 mark

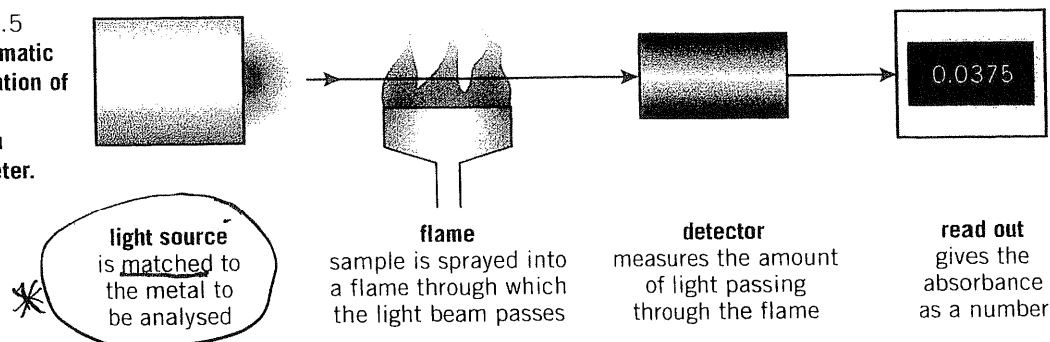
Question 18 (5 marks)

Atomic absorption spectroscopy (AAS) is an extremely useful tool in the detection of metal ion concentrations.

(a) Explain why AAS is of little use in identifying unknown substances.

3

Figure 11.5
A diagrammatic
representation of
an atomic
absorption
spectrometer.



18 (a)

Atoms absorb & emit characteristic light frequencies.
AAS [light source] emits light of specific frequency of (metallic) element under test.
If sample does not contain the same element as in the AAS light source, then it is very unlikely that light will be absorbed in the AAS detector and the 'read out' to be meaningful.
AAS can't be used for an unknown substance.
little use in identifying non-metallic compounds.

Coherent explanation

Reasonable explanation

Some background on AAS

Marks

3

2

1

18 (b).

AAS may be used to detect cations in very low concns - ppm to ppb.

This technique can measure trace elements, eg Co. Mn beneficial to crops + animals in very low concns.

Marks

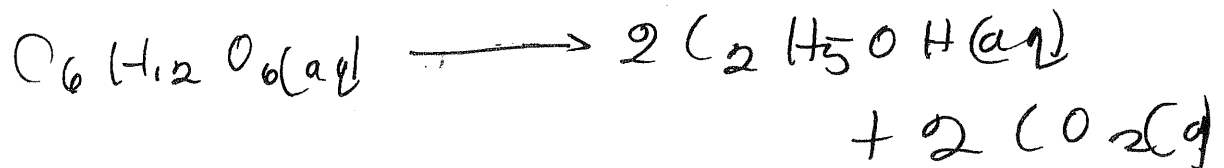
- must mention detection at very low concn - ppm or ppb

Give an example of the use of AAS mentioning a trace metal by name.

Question 19 (6 marks)

The fermentation of glucose is a chemical process which has been known to humans for at least 5 thousand years.

- (a) Write a chemical equation to represent the fermentation of glucose.



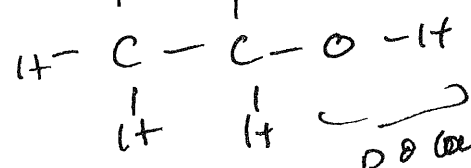
Equation must include correct states
for 2 marks
(Balanced eqn, but incorrect states: 1 mark)

- (b) Under what physical conditions is fermentation optimised?

(b) Anaerobic or low O_2
Temp. - body temp, ~ 37°C
or even 'room temp' } both needed for 1 mark.
Moderate temp' - not accepted.

(c) One of the products of the fermentation process is frequently used as a solvent for both polar and non-polar solutes. Account for ethanol's ability to do this.

(c) Ethanol



small non-polar / g.p.

polar end. enables ethanol to form H-bond bonds and dipole-dipole forces with polar molecules. eg H_2O , glucose

enables ethanol to form dispersion forces with non-polar substances eg I_2 , hydrocarbons.

Good account - including polar + non-polar sections of ethanol molecule
- types of i/m forces possible with polar + non-polar ends of molecule.
- egs of polar + non-polar solutes

Marks

3

Reasonable account 2

Some description of ethanol as a solvent - eg like dissolves like 1

Section I – Part B (continued)

CR1B

Class

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Student Number

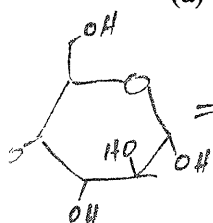
Marks

Question 20 (5 marks)

Fossil fuels, which at present make up the bulk of the raw material used in the plastics industry, are a finite resource and likely to become severely depleted in the near future. Biopolymers have been suggested as a possible replacement for the petrochemicals produced from fossil fuels.

- (a) Cellulose is often considered the most useful compound from which to produce biopolymers. Describe the structure of cellulose.

2



= A { Cellulose is a polymere chain made from repeating glucose monomers, where every second monomer is inverted. ...A-U-A-U-A...

- (b) (i) Identify a biopolymer which has recently been developed or is in the process of being developed, for commercial use.

1

(eg) PHA (or another bio polymer)
 (NB: ethanal is not a polymer!)

- (ii) Name the specific enzyme or organism used to synthesise this biopolymer.

1

(eg) Alcaligenes eutrophes

- (c) Suggest one benefit (apart from their renewability), of using biomass to produce polymers.

1

or • Biopolymers generally are biodegradable
 or • Conservation of crude oil for use as fuel
 or • (+ others)

Marks

Question 21 (3 marks)

(a) Draw electron dot diagrams to show:

(i) an oxygen molecule.

1

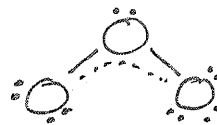


Generally well done

(ii) an ozone molecule. (Must be bent!)

1

Is really a resonance structure - not in syllabus
 ∴ had to accept "incorrect" octet version.



(b) (* Do NOT use O_3 as an example of a co-ordinate covalent bond - use NH_4^+ or H_3O^+)
 State the difference in stability of ozone gas and oxygen gas.

O_3	O_2
unstable	stable
or (+ others)	

Marks

Question 22 (3 marks)

Esters are produced by reaction of an alkanoic acid and an alcohol.

- (a) Name a straight-chained alkanoic acid.

1

Propanoic acid (or other)

- (b) Name a primary alcohol.

1

Ethanol (had to say 1-propanol etc if used C₃ or higher)

- (c) Name the ester that would be produced by refluxing this acid with this alcohol.

1

Ethyl propanoate (must be correct using (a) & (b))

Marks

Question 23 (7 marks)

Over time, the definitions of acids and bases have been refined. Using the historical development of ideas about acids, evaluate how advances in scientific understanding changed the direction of scientific thinking.

7

~1776 Lavoisier } NB contains oxygen $\neq$ contains O_2 !!!
 ~1810 Davy }
 ~1884 Arrhenius } all in syllabus (p55)
 ~1923 Bronsted/Lowry }

"Over time" + "historical development" mean that the order was important!

MARKING GUIDELINES

(See syllabus p 55)

<u>Part-A</u>	All 4 people (above) <u>named</u> - including their ideas - must be accurate & "in time"	(5)
	Omitted one person <u>or</u> error of idea	(4)
	Omitted two people <u>or</u> several errors <u>or</u> omitted one person <u>AND</u> an error	(3)
	Omitted two people <u>AND</u> several errors	(2)
	One person & idea correct	(1)

Part B EVALUATE - make a judgement based on
 - determine the value of....

Excellent evaluation	(2)
Good evaluation	(1)

NB Did not accept a 'restatement of the question' as an evaluation.

Section I – Part B (continued)

CRIB

AKBB

Class

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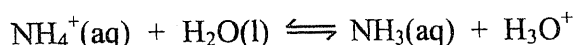
Student Number

Marks

Question 24 (5 marks)

A student was investigating the acid/base nature of salts, by adding the dry solid salts one at a time to water and then testing their pH.

When he did this with ammonium chloride, he noted that the $\text{pH} < 7$, and assumed that the following action had occurred.



- (a) Why does the above equation illustrate a Brønsted-Lowry acid, rather than an Arrhenius acid? 2

Ans. Arrhenius acid - ionises in water to produce H^+ ions. The above system does not. A Brønsted-Lowry acid - is defined as a proton donor. Both NH_4^+ & H_3O^+ donate a proton. ①

- (b) From the above equation, give one example of an acid and its conjugate base, respectively. 1

NH_4^+ & NH_3 OR H_3O^+ & H_2O

- (c) Briefly outline how you would perform a first-hand investigation to determine the concentration of an acidic substance using a computer-based technology. 2

Accurate description of hardware (i.e. pH meter) ①
Conversion of raw data into $[\text{H}^+]$ & then acid conc. ①

Marks

Question 25 (3 marks)

- (a) Identify two metallic ions which are found in hard water.

1

$Mg^{2+} + Ca^{2+}$

- (b) Describe a simple method of determining the hardness of water in a school laboratory.

2

* Titrating (brief description) - 1 MARK
with EDTA - 1 MARK

OR

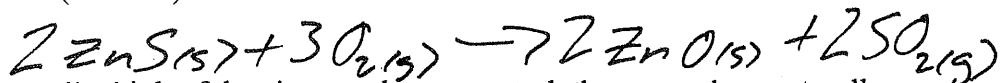
* Add soap & shake - 1 MARK
Description of method including comparison
to other / standard - 1 MARK.

OR

* Any alternative that would work
+ description

Marks

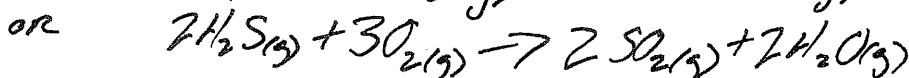
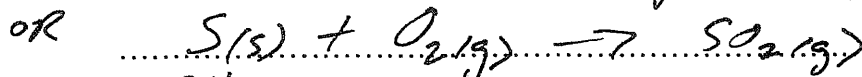
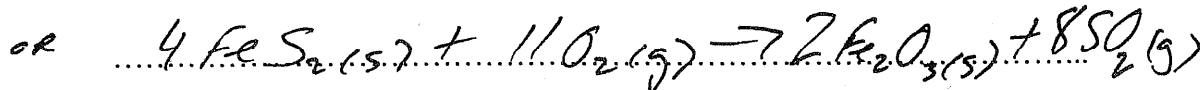
Question 26 (6 marks)



While we usually think of the air around us as neutral, the atmosphere naturally contains acidic oxides of carbon, nitrogen and sulfur.

- (a) (i) Describe, using an equation, an example of a chemical reaction which releases sulfur dioxide.

2



Perfect - 2 MARKS
Any mistake (-1)

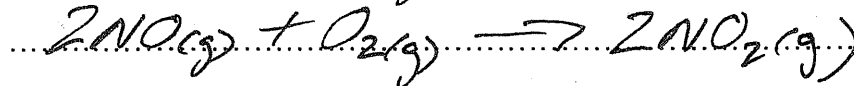
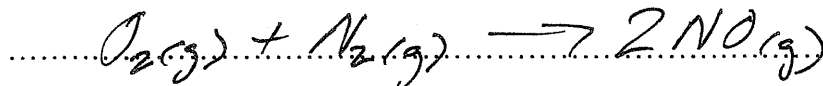
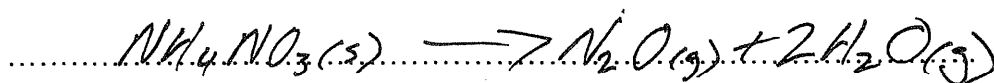
- (ii) Identify a natural source of sulfur dioxide.

1

Volcanic eruptions OR Bush fires OR Decaying organic matter

- (b) (i) Describe, using an equation, an example of a chemical reaction which releases an oxide of nitrogen.

2



Perfect - 2 MARKS
Any mistake (-1)

- (ii) Identify a natural source of nitric oxide (NO), a gas that is capable of destroying ozone, and is involved in the production of photochemical smog.

1

lightning

Section I – Part B (continued)

Class

Student Number

Marks

Question 27 (4 marks)

* many confused n/p with p/n

(a) Discuss the conditions under which nuclei are stable.

(SEE p102)

2

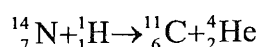
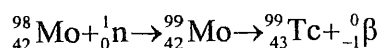
neutron-to-proton ratio lies within narrow limits ①
 too massive ($Z > 82$)

OR n/p 1.0 for first 20 then increases to ~1.3 } ①

OR n/p too low K-capt/ β^+ emits n/p too high β^- decay

(b) The two equations below represent the formation of significant artificial isotopes:

2



Tc-99 is the most widely used radioactive isotope for diagnostic studies in nuclear medicine. C-11 is incorporated into organic compounds and used as a tracer in positron emission tomography (PET).

Discuss the production of commercial isotopes using these and / or other relevant examples.

NB
not
"use"

neutron-rich isotopes like Tc-99 made in a nuclear reactor ①

neutron-poor isotopes like C-11 made in a cyclotron ①

(it is possible to make Tc-99 in a cyclotron but NOT by the process shown above)

* SEE pp 108/9

Marks

Question 28 (6 marks)

Galvanic cells were constructed using the metals A – E and the voltages measured under standard conditions. The results are shown in the table below.

Cell reaction	$E^\circ_{\text{cell}} / \text{V}$
$\text{A} + \text{B}^{2+} \rightarrow \text{A}^{2+} + \text{B}$	0.98
$\text{B} + \text{D}^{2+} \rightarrow \text{B}^{2+} + \text{D}$	1.05
$2\text{C} + \text{B}^{2+} \rightarrow 2\text{C}^+ + \text{B}$	1.68
$\text{B} + \text{B}^{2+} \rightarrow \text{B}^{2+} + \text{B}$	0.00
$\text{B} + \text{E}^{2+} \rightarrow \text{B}^{2+} + \text{E}$	0.66

- (a) Draw a labelled diagram of one of the cells used and identify clearly the reference cell.

3

ALMOST NOBODY
DID THIS!!

DIAG: $2 \times \frac{1}{2}$ cell

1 x salt bridge

~~NEARLY DRAWN~~ NEARLY DRAWN & CORRECTLY DRAWN (SO COME)

LABEL: $\text{M} / \text{M}^{2+} \times 2$

salt bridge

e^- flow OR cathode/anode

ID THE REF CELL B / B^{2+} cell

- (b) Explain what is meant by standard conditions.

1

1 atm (or 1 bar); 25°C (or 298K); all concs 1 M
(all 3 required)

Question 28 continued on page 21

Marks

Question 28 (continued)

- (c) Construct a table of standard (half-cell) potentials from the data collected.

1

$C^+ + e^- \rightleftharpoons C$	E_{red}^0/V -1.68
$A^{2+} + 2e^- \rightleftharpoons A$	-0.98
$B^{2+} + 2e^- \rightleftharpoons B$	0.00
$E^{2+} + 2e^- \rightleftharpoons E$	+0.66
$D^{2+} + 2e^- \rightleftharpoons D$	+1.05

All

(1)

- (d) (i) Identify the best reducing agent.

1

C (the most active metal)

- (ii) Identify the best oxidising agent.

D²⁺ (ion of the least active metal)BOTH
(1)⊗ METALS cannot be oxidising agents

Marks

Question 29 (8 marks)

Polyethylene is a chemical which has been of significant commercial importance in the past fifty years.

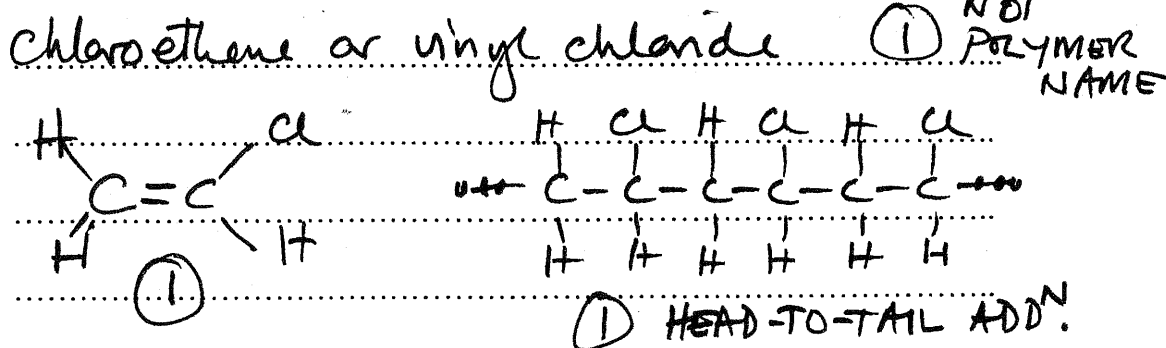
- (a) Outline the major steps in the industrial production of polyethylene, from the raw material used, to the finished product. 3

FRACTIONAL DISTILLATION OF PETROLEUM / CRUDE OIL (1)
 (CATALYTIC) CRACKING OF HIGHER FRACTIONS → ETHENE (1)
 OR (ADDITION POLYMERISATION OF ETHENE (1)
 POLY. in pres. & catalyst & / or high T, P)

Answers often contained much INCORRECT (or poorly expressed) detail.

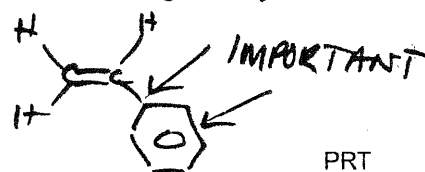
- (b) Many commercial polymers are produced by the modification of ethene molecules, such that a hydrogen is replaced by a side group, followed by a polymerisation reaction.

- (i) Identify one such "modified ethene" monomer, either by its common or systematic name, and using complete structural formula, write an equation to represent the polymerisation reaction, using three monomer units. 3



MANY who chose styrene / phenylethene couldn't draw a benzene ring (phenyl group) or represent its connection to the vinyl group

Question 29 continued on page 23



Marks

Question 29 (continued)

N.B.

- (ii) Describe a use for the polymer you have identified, in part (i), in terms of its physical or chemical properties.

2

① PROPERTY

① USE CONNECTED TO PROPERTY

⊗ REF/SOURCE pp 25/6

N.B. IF the property is the result or modification of the polymer by additives, formation of a foam etc, this had to be described explicitly in the answer. Thus, for example "PVC is flexible" is incorrect (see p 25)

BLANK PAGE

The Chemistry of Art.

a(i) Arsenic(III) sulfide (1)

(ii) Pigments can be inhaled as dust
 ingested (from hands)
 absorbed through cuts

(1) any two
 OR
 (1)

Pb/Hg compounds are toxic } affect CNS
 As/Sb compounds are toxic } body treats them like P (1)

www.conmex.com/sub/artpage/ART/ARTHmedieval.html

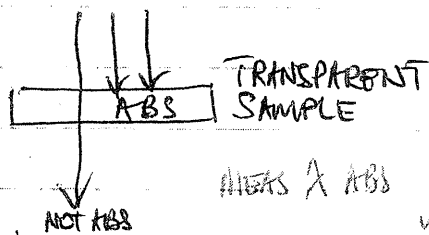
(b) (i) A paint consists of a pigment (1)
 and a medium (binder)

(ii) The support is prepared by a layer of gesso
 The paint is applied as egg tempera } pigment
 mixed with egg yolk (1) NB NOT TEMPERA

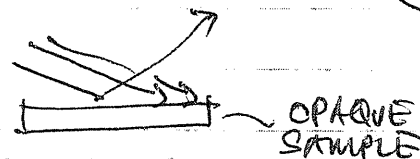
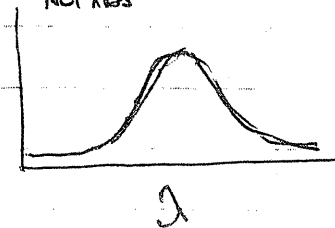
"St John the Baptist with St John the Evangelist & St James" (1)

SPECTRUM (S)
 SPECTRA (PL)
 (c)

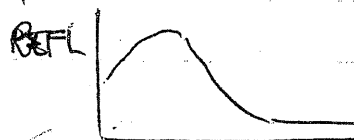
NOT ATOMIC
 SPECTRA
 (1) ABS



MEAS X ABS



MEAS X NOT ABS



"Complementarity"

(1) A plot of absorption intensity
 (amt of light abs) against wavelength

A plot of intensity
 of refl. light against
 wavelength.

(1) Absorption associated with changes in energy levels,

CHEM OF ART

2

- (d) Sharp, discrete lines
 ABS black lines on coloured background
 EMI Coloured lines on black background.
- ① STATE FEATURES
 Freq / λ of abs & em lines are the same.

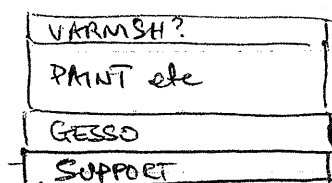
- ① E emitted when e^- move from higher E-level to a lower E-level
 E abs when e^- move from lower E-level to a higher E-level

① Atoms with many electrons have more E-levels and more complex spectra than H.

① EXPLAIN FEATURES
 must refer to ~~spectra~~ given (say which is which in diagram)

Q "EXPLAIN"

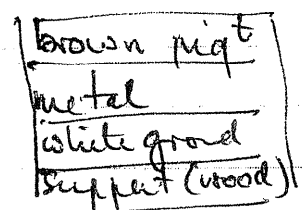
(e)



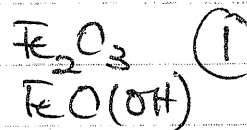
Pulses dig down



①
 ATOMS ID BY
 ATOMIC emission spectra



5th pulse brown pigment Fe present
 (Ca & Na don't form coloured compounds)



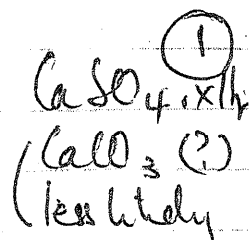
10th pulse metallic layer
 (Ca is a very reactive metal)

Ag present



14th pulse white ground

Ca present



30

CRIB-AKBB

i eg Rubber - 1 MARK.

ii eg Styrene butadiene or neoprene - 1 MARK.

iii eg historical context WWII Japan/America.

OR cost/efficiency

OR no need for land clearance etc 1 MARK.

OR. Simply stating Demand was not sufficient.

bi Any specific example of H_2SO_4 as a dehydrating agent. 1 MARK.

Accepted this time { H_2SO_4 acting as a catalyst in dehyd. rxn }
but NOT GOOD examples { OR dehydrating system in EM. }

ii Conc H_2SO_4 plus example of oxidation - 1 MARK

Equation to represent this oxidation process - 1 MARK.

OR Simple redox oxidation equation - 1 MARK

$$c. \quad K = \frac{[CO][Cl_2]}{[COCl_2]}$$

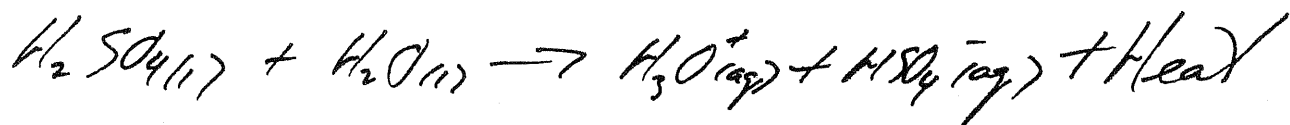
identification of oxidised species 1 MARK

- 1 MARK.

$$[Cl_2] = 0.08M, [CO] = 0.08M, [COCl_2] = 0.02M \quad 1-MARK$$

$$K = \frac{0.08^2}{0.02} = \underline{\underline{0.32}} \quad 1 MARK$$

d.



* extremely exothermic therefore may sp. V

Essential to add acid to water ^{slowly/stirring} rather than vice versa so at worst only dilute acid will sp. V. 1-MARK

* $C_1V_1 = C_2V_2$ or actual calculation 1-MARK
or H_2SO_4 1 : 8 H_2O

* Must wear safety specs, gloves and protective clothing. 1-MARK.

OR if reasonable attempt at 2 or more of above points - 1 MARK

e. One mark does NOT equal one mark!!

1 use mentioned - 0 MARKS
2 uses mentioned - 1 MARK
3 uses mentioned - 2 MARKS

PLUS

Details or equation of 1 use - 1 MARK
Details or equations of 2 uses - 2 MARKS.

NB- The production of ethene from ethanol is not currently a significant industrial process